Notes: Olken (JPE, 2007)

Monitoring Corruption: Evidence from a Field Experiment in Indonesia

Contents

1 Big picture

- **Question.** Can corruption in local public works projects be reduced by top-down monitoring, grassroots participation, or both? The paper asks this in a setting where corruption is directly measured rather than inferred from perceptions.
- **Setting.** The experiment studies 608 Indonesian villages in East Java and Central Java that were about to build village road projects under the Kecamatan Development Project (KDP), a nationwide village infrastructure program funded through a World Bank loan.
- **Core policy contrast.** The paper compares two broad anticorruption strategies: traditional external audits by the government audit agency and community participation through village accountability meetings.
- **Audit treatment.** Some villages were randomly told that their road project would be audited by the central government audit agency. This increased the probability of audit from roughly 4% to essentially 100%.
- **Grassroots treatments.** Other villages received hundreds of invitations to accountability meetings, and a subset also received anonymous comment forms that allowed villagers to report concerns without publicly identifying themselves.
- **Measurement contribution.** The paper measures corruption by comparing official reported project costs with independent engineering estimates of actual road costs. Engineers dug core samples, measured materials, surveyed prices, and estimated labor inputs.
- **Main result.** Government audits reduced missing expenditures by about eight percentage points. Given control-village missing expenditures of roughly 28 percentage points for major road items, this is a large reduction.
- **Grassroots result.** Participation had little average effect on total missing expenditures, though invitations reduced missing labor expenditures and comment forms worked better when distributed through schools rather than village officials.
- **Mechanism.** Audits reduced corruption even in a highly corrupt environment, but they did not eliminate it. Most audit findings were procedural, not direct proof of criminal theft, so sanctions remained imperfect.
- **Economic takeaway.** Top-down monitoring can work. Grassroots monitoring can also help, but mainly where free-riding and elite capture are limited.

2 Incremental contribution of the paper

- **From perceived corruption to measured corruption.** Much corruption research relies on survey perceptions. This paper constructs a direct objective measure of missing expenditures by comparing reported costs with independent engineering estimates.
- **A randomized field experiment on corruption.** The paper brings experimental identification to a question that is usually studied with cross-sectional or observational evidence.
- **Top-down versus bottom-up comparison.** The same setting is used to compare government audits and community monitoring, two leading policy approaches to corruption control.

- **Mechanism-level evidence.** The paper does not only estimate average treatment effects. It separately studies prices versus quantities, materials versus labor, meeting participation, meeting discussion, auditor findings, and nepotistic employment.
- **Public-good logic of monitoring.** The results clarify when grassroots monitoring is likely to work. Laborers have private incentives to monitor missing wages, but the whole village faces a free-rider problem in monitoring road materials.
- **Elite-capture contribution.** The paper shows that comment forms reduce missing expenditures when distributed through schools, which bypass village officials, but not when distributed through neighborhood heads.
- **Substitution across forms of corruption.** Audits reduce missing expenditures but appear to increase the probability that project jobs go to relatives of village or project officials. This suggests that corruption can shift toward less-audited margins.
- **Policy-cost evidence.** The cost-benefit analysis suggests that the audit treatment was cost-effective even when accounting for audit costs, villagers' time costs, and distributional weights.
- **Broader contribution.** The paper provides one of the clearest field-experimental demonstrations that classic monitoring-and-punishment logic can reduce corruption, while also showing why community monitoring is not a universal substitute for professional auditing.

3 Data and measurement

3.1 Program setting

- The Kecamatan Development Project gave participating subdistricts annual block grants for village-level projects.
- Villages proposed small infrastructure projects and sometimes microcredit seed capital. The analysis focuses on nonasphalt village road projects to keep the engineering measurement comparable.
- A typical funded village received about Rp. 80 million, or roughly US$8,800, for infrastructure. This was large relative to ordinary village budgets.
- Village implementation teams procured materials, hired labor, and built the project without contractors.
- Accountability meetings were required before later funding tranches were released. These meetings were open in principle but usually attended by a small village elite.

Interpretation: The setting is well suited for a corruption experiment because projects are economically meaningful, locally managed, and physically measurable after completion.

3.2 Experimental treatments

- **Audit treatment.** Villages were told before construction that the central government audit agency would audit their project with probability one.

- **Invitations treatment.** Villages received 300 or 500 written invitations to accountability meetings, distributed broadly to reduce the village head's control over meeting attendance.
- **Invitations plus comment forms.** Villagers also received anonymous comment forms that could be returned in sealed drop boxes. Summaries were read at accountability meetings.
- Audit randomization was clustered by subdistrict to avoid spillovers. Participation treatments were randomized village by village.
- Treatments were announced after funds and project designs were finalized but before procurement and construction began.

Interpretation: The treatment timing matters. Villages could not change project type or planned budget in response to treatment assignment, so post-treatment differences can be interpreted as changes in implementation behavior.

3.3 Measuring missing expenditures

The main dependent variable is the difference between what the village reported spending and what an independent engineering survey estimated the project actually cost. Conceptually,

$$\text{Percent Missing} = \log(\text{Reported Cost}) - \log(\text{Estimated Actual Cost}). \tag{1}$$

- Reported expenditures come from village accountability reports.
- Actual quantities of road materials were estimated by engineers using core samples and physical measurements.
- Material prices were estimated from subdistrict-level supplier and buyer surveys.
- Labor costs were estimated using worker surveys, task quantities, and worker-capacity assumptions.
- Calibration roads were built to estimate normal material loss ratios and measurement losses.

Interpretation: The measure is not perfect, but it is much closer to actual corruption than perceptions. It captures missing money after adjusting for normal construction and measurement losses.

3.4 Meeting and household data

- Enumerators attended accountability meetings and recorded attendance, who spoke, what issues were discussed, and whether serious actions were taken.
- The household survey measured participation in the project, perceptions, social connections, family ties to officials, and whether household members worked on the project.
- Auditor reports provide information on procedural and physical findings from the official audits.

Interpretation: These supplementary data allow the paper to move beyond the main corruption outcome and study mechanisms: participation, problem discussion, audit quality, and substitution into nepotism.

4 Models and empirical specifications

4.1 Treatment-effects equation

The main estimating equation is an OLS regression of missing expenditures on treatment indicators:

$$\text{PercentMissing}_{ijk} = \alpha + \beta_1 \text{Audit}_{jk} + \beta_2 \text{Invitations}_{ijk} + \beta_3 \text{InvitationsComments}_{ijk} + \varepsilon_{ijk}, \quad (2)$$

where i indexes villages, j indexes subdistricts, and k indexes audit strata.

- Audit assignment is at the subdistrict level, so standard errors allow clustering by subdistrict.
- Participation treatments are randomized at the village level.
- Specifications are reported with no fixed effects, engineer-team fixed effects, and stratum fixed effects.
- The dependent variable is measured for major road items, road plus ancillary projects, materials only, and unskilled labor only.

Interpretation: Because treatment assignment is randomized, the coefficients estimate causal effects of monitoring treatments on missing expenditures.

4.2 Audit decomposition

The paper decomposes missing expenditures into price and quantity components.

- If corruption occurs through inflated receipts for materials or labor, it can appear as inflated prices, inflated quantities, or both.
- Commodity prices are easier for monitors to verify than true quantities delivered and embedded in the road.
- The audit results show that reductions in missing expenditures are driven mainly by quantities, not prices.

Interpretation: The decomposition supports the corruption interpretation. Officials did not mainly overstate prices; they overstated quantities of materials or labor.

4.3 Auditor findings

The paper compares official auditor scores with the independent engineering survey.

$$\text{EngineeringScore}_{ij} = \alpha_j + \theta_1 \text{AuditorPhysicalScore}_{ij} + \theta_2 \text{AuditorAdminScore}_{ij} + u_{ij}. \quad (3)$$

- Auditor physical scores are positively related to engineering-team physical scores.
- Auditor administrative scores are negatively related to missing expenditures.

- However, most official audit findings are procedural: missing documentation, bad ledgers, or improper procurement procedures.

Interpretation: Audits contain real information, but the evidence often does not directly prove theft. This helps explain why audits reduce but do not eliminate missing expenditures.

4.4 Nepotism equation

The household-level nepotism analysis estimates whether relatives of officials are more likely to work on the project in audited villages:

$$\text{Worked}_{hijk} = \gamma_k + \delta_1 \text{Audit}_{jk} + \delta_2 \text{Family}_{hijk} + \delta_3 (\text{Audit}_{jk} \times \text{Family}_{hijk}) + X'_{hijk}\Lambda + e_{hijk}. \quad (4)$$

- The outcome is whether a household member worked for pay on the project.
- Family indicators identify relatives of village government officials or of the project head.
- The interaction term measures whether audits change the relative probability that official families get paid project jobs.

Interpretation: The results suggest possible substitution. When missing expenditures become riskier, officials may shift toward job allocation to relatives, a margin less directly targeted by audits.

4.5 Participation mechanisms

For participation treatments, the paper separately estimates effects on meeting attendance, active speaking, corruption-related discussion, serious responses, and missing expenditures.

- Invitations are a strong first stage for attendance and nonelite attendance.
- Both invitations and comment forms increase the chance that corruption-related problems are discussed.
- Comment forms increase the probability of a serious response at the meeting, but the absolute magnitude is small.
- Participation treatments have little average effect on total missing expenditures but reduce missing labor expenditures in some specifications.

Interpretation: Participation changed behavior at meetings, but not enough to strongly reduce overall corruption because materials made up most project spending and were harder to monitor through grassroots incentives.

5 Identification and empirical design

5.1 Randomization and balance

- Audit assignment was randomized by subdistrict. Participation treatments were randomized by village.

- Table 2 shows that baseline village characteristics are not jointly significant predictors of treatment assignment.
- Project designs and budgets were finalized before treatments were announced.
- The engineering survey was independent of the official audit process.

Interpretation: The design is unusually strong for a corruption study. Random assignment plus direct measurement gives a clean estimate of treatment effects on missing expenditures.

5.2 Threats to interpretation

- **Measurement error.** Engineering estimates can be noisy, especially because normal material loss must be calibrated. The paper uses test roads and log differences to reduce this concern.
- **Builder competence versus corruption.** Audits might improve construction competence rather than reduce theft. The paper examines low-cost quality measures and finds that they do not explain the missing-expenditure results.
- **Accounting changes.** Audits might change reporting rather than actual spending. The paper compares actual and reported expenditures relative to pre-treatment plans and argues that the evidence points to changes in actual expenditures.
- **Audit spillovers.** Audit treatment is clustered by subdistrict to reduce within-subdistrict spillovers. Cross-subdistrict spillovers appear limited.
- **Substitution.** Audits can shift corruption to less-monitored margins, such as nepotistic hiring.

Interpretation: The paper is careful about mechanisms and measurement. The main audit effect is credible, but it should be interpreted as a reduction in measured missing expenditures, not complete elimination of all corrupt behavior.

5.3 What the design can and cannot identify

- The design identifies short-run effects of monitoring treatments on village road projects.
- It does not fully identify long-run equilibrium responses, such as whether villages learn to bribe auditors or whether candidate selection into village offices changes.
- It identifies effects for locally managed infrastructure projects, especially public-good projects with materials and labor inputs.
- It does not imply that grassroots monitoring is ineffective in all settings. The paper explicitly suggests it may work better for private goods such as subsidized food, medical services, or school benefits, where beneficiaries have stronger private incentives to monitor.

Interpretation: The external-validity message is conditional. Top-down audits appear powerful for public infrastructure; grassroots monitoring is likely to depend strongly on information, incentives, and capture.

6 Tables and figures

6.1 Table 1 and Table 2: treatment cells and randomization balance

Read:

- Table 1 shows the experiment's cross-randomized design: control, invitations, and invitations plus comment forms, each crossed with audit and no-audit status.
- The total sample is 608 villages. Of these, 283 villages are in audit subdistricts and 325 are in no-audit subdistricts.
- Table 2 checks whether treatment assignment predicts baseline village characteristics such as population, budget, poverty, distance to subdistrict, and village-head characteristics.
- The joint p-values are 0.18 for audits, 0.92 for invitations, and 0.52 for comment forms conditional on invitations.

Economic message: The experimental design creates comparable treatment and control groups. This is the foundation for interpreting later differences in missing expenditures as causal effects of monitoring.

TABLE 1
Number of Villages in Each Treatment Category

	Control	Invitations	Invitations Plus Comment Forms	Total
Control	114	105	106	325
Audit	93	94	96	283
Total	207	199	202	608

Note.—Tabulations are taken from results of the randomization. Each subdistrict faced a 48 percent chance of being randomized into the audit treatment. Each village faced a 33 percent chance of being randomized into the invitations treatment and a 33 percent chance of being randomized into the invitations plus comment forms treatment. The randomization into audits was independent of the randomization into invitations or invitations plus comment forms.

(a) Table 1: treatment categories

TABLE 2
Relationship between Treatments and Village Characteristics

	Audits (1)	Invitations (2)	Comments (Conditional on Invitations) (3)
Village population (000s)	−.007	.004	.001
	(.012)	(.007)	(.009)
Mosques per 1,000	−.018	.000	.012
	(.038)	(.024)	(.028)
Total budget (Rp. millions)	−.001	−.000	−.000
	(.001)	(.000)	(.000)
Number subprojects	−.017	.002	−.017
	(.025)	(.013)	(.016)
Percent households poor	.246*	.069	.033
	(.126)	(.080)	(.111)
Distance to subdistrict	−.001	−.002	.001
	(.005)	(.004)	(.005)
Village head education	.012	−.002	.016
	(.009)	(.007)	(.010)
Village head age	.004	.003	−.000
	(.003)	(.003)	(.003)
Village head salary (hectares)	.011*	.004	.006
	(.007)	(.003)	(.004)
Mountainous dummy	.134*	.010	.077
	(.074)	(.037)	(.049)
Observations	577	577	381
p-value of all listed variables	.18	.92	.52

Note.—Results reported are marginal effects from Probit regressions. Robust standard errors are in parentheses, adjusted for clustering at the subdistrict level.

(b) Table 2: balance across village characteristics

6.2 Table 3 and Table 4: baseline missing expenditures and audit effects

Read:

- Table 3 shows that the average project is about US$8,875 and that roads account for 76.6% of reported expenses.
- Rocks are the largest reported road input, accounting for 48.4% of road expenses. Unskilled labor accounts for 19.6%.
- The mean percent missing is 23.7% for major road items and 24.7% for road plus ancillary projects.

- Table 4 shows that audits reduce missing expenditures for major road items by 8.5 percentage points without fixed effects and 7.6 percentage points with engineer fixed effects.
- For roads plus ancillary projects, the audit effect is about 8.6 to 9.1 percentage points and is statistically significant across the main specifications.

Economic message: Missing expenditures are economically large, and audits produce a large reduction. The estimates imply roughly a 30% reduction relative to the missing-expenditure level in control villages.

TABLE 3
SUMMARY STATISTICS

	Summary Statistics
Total project size (US$)	8,875 (4,401)
Share of total reported expenses:	
Road project	.766 (.230)
Ancillary projects (culverts, retaining walls, etc.)	.154 (.181)
Other projects (schools, bridges, irrigation, etc.)	.079 (.166)
Share of reported road expenses:	
Sand	.099 (.080)
Rocks	.484 (.143)
Gravel	.116 (.181)
Unskilled labor	.196 (.125)
Other	.105 (.164)
Percent missing:	
Major items in road project	.237 (.343)
Major items in roads and ancillary projects	.247 (.350)
Materials in road project	.203 (.395)
Unskilled labor in road project	.273 (.851)
Observations	538

NOTE.—Statistics shown are means, with standard deviations in parentheses. Data on expenditures are taken from the 538 villages for which percent missing in road and ancillary projects could be calculated. Exchange rate is Rp. 9,000 = US$1.00.

(a) Table 3: summary statistics

TABLE 4
AUDITS: MAIN THEFT RESULTS

			No Fixed Effects		Engineer Fixed Effects		Stratum Fixed Effects	
Percent Missing[a]	Control Mean (1)	Treatment Mean: Audits (2)	Audit Effect (3)	p-Value (4)	Audit Effect (5)	p-Value (6)	Audit Effect (7)	p-Value (8)
Major items in roads (N = 477)	.277	.192	−.085*	.058	−.076**	.039	−.048	.123
	(.033)	(.029)	(.044)		(.036)		(.031)	
Major items in roads and ancillary projects	.291	.199	−.091**	.034	−.086**	.022	−.090***	.008
(N = 538)	(.030)	(.030)	(.043)		(.037)		(.034)	
Breakdown of roads:								
Materials	.240	.162	−.078	.143	−.063	.136	−.034	.372
	(.038)	(.036)	(.053)		(.042)		(.037)	
Unskilled labor	.312	.231	−.077	.477	−.090	.304	−.041	.567
	(.080)	(.072)	(.108)		(.087)		(.072)	

NOTE.—Audit effect, standard errors, and p-values are computed by estimating eq. (1), a regression of the dependent variable on a dummy for audit treatment, invitations treatment, and invitations plus comment forms treatments. Robust standard errors are in parentheses, allowing for clustering by subdistrict (to account for clustering of treatment by subdistrict). Each audit effect, standard error, and accompanying p-value is taken from a separate regression. Each row shows a different dependent variable, shown at left. All dependent variables are the log of the value reported by the village less the log of the estimated actual value, which is approximately equal to the percent missing. Villages are included in each row only if there was positive reported expenditures for the dependent variable listed in that row.
[a] Percent missing equals log reported value − log actual value.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(b) Table 4: audit treatment effects

6.3 Figure 1 and Table 5: distributional evidence and price-versus-quantity decomposition

Read:

- Figure 1 plots the empirical CDF and estimated PDF of missing expenditures for audit and control villages.
- The audit distribution is shifted toward lower missing expenditures, indicating that the audit effect is not driven only by a few outliers.
- Table 5 decomposes missing expenditures into reported-versus-actual prices and reported-versus-actual quantities.
- In control villages, prices are not meaningfully inflated on average, while quantities are substantially missing.
- The audit effect on prices is essentially zero, while the quantity effect is negative, around 6.9 percentage points in the no-fixed-effects specification.

Economic message: The main corruption margin is quantity inflation rather than price inflation. This fits the idea that quantities embedded in the road are harder for ordinary monitors to verify than commodity prices.

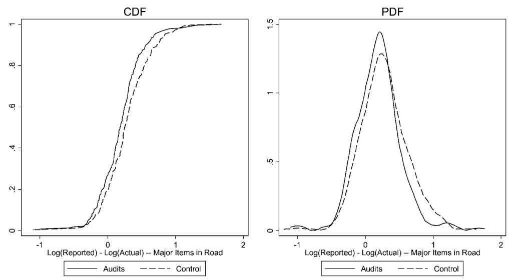

Fig. 1.—Empirical distribution of missing expenditures. The left-hand figure shows the empirical CDF of missing expenditures for the major items in a road project, separately for villages in the audit treatment group (solid line) and the control group (dashed line). The right-hand figure shows estimated PDFs of missing expenditures for both groups; PDFs are estimated using kernel density regressions using an Epanechnikov kernel.

(a) Figure 1: distribution of missing expenditures

TABLE 5
Audits: Prices vs. Quantities

			No Fixed Effects		Engineer Fixed Effects		Stratum Fixed Effects	
Percent Missing[a]	Control Mean (1)	Treatment Mean: Audits (2)	Audit Effect (3)	p-Value (4)	Audit Effect (5)	p-Value (6)	Audit Effect (7)	p-Value (8)
Prices (N = 494)	−.018	−.016	.002	.949	.011	.662	.018	.434
	(.017)	(.021)	(.027)		(.025)		(.023)	
Quantities (N = 477)	.276	.207	−.069*	.092	−.069*	.051	−.048	.120
	(.030)	(.028)	(.041)		(.035)		(.031)	

Note.—See the note to table 4. Reported and actual prices are defined as a weighted average across different commodities (rock, sand, gravel, and unskilled labor), where each commodity is weighted by the reported quantity. Reported and actual quantities are also defined as a weighted average, where each commodity is weighted by the reported prices.
[a] Percent missing equals log reported value − log actual value.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(b) Table 5: prices versus quantities

6.4 Table 6 and Table 7: what auditors found

Read:

- Table 6 compares BPKP auditor scores with the independent engineering-team scores.
- Auditor physical scores are positively correlated with engineering-team physical scores.
- Auditor administrative scores are negatively correlated with missing expenditures. A better administrative score is associated with less missing money.
- Table 7 shows that auditors found some issue in 90% of audited villages.
- Most findings were procedural: 50% had daily expenditure ledgers not in accordance with procedures, 38% had procurement problems, and 28% had insufficient documentation of material receipts.

Economic message: Audits contain useful information, but official findings often document weak procedures rather than direct evidence of theft. This explains why the threat of audit matters but does not drive corruption to zero.

TABLE 6
Relationship between Auditor Findings and Survey Team Findings

	Engineering Team Physical Score (1)	Engineering Team Administrative Score (2)	Percent Missing in Road Project (3)
Auditor physical score	.109**	−.067	.024
	(.043)	(.071)	(.033)
Auditor administrative score	.007	.272**	−.055**
	(.049)	(.133)	(.027)
Subdistrict fixed effects	Yes	Yes	Yes
Observations	248	249	212
R^2	.83	.78	.46

Note.—Robust standard errors are in parentheses, adjusted for clustering at subdistrict level. Auditor scores refer to the results from the final BPKP audits; engineering team scores refer to the results from the engineering team that was sent to estimate missing expenditures. The results from the engineering team were not shared with the BPKP audit team. All specifications include subdistrict fixed effects, which therefore hold constant both the BPKP audit teams and the engineering teams. For both physical and administrative scores, scores are normalized to have mean zero and standard deviation one.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(a) Table 6: auditor findings and independent survey findings

TABLE 7
Audit Findings

	Percentage of Villages with Finding
Any finding by BPKP auditors	90%
Any finding involving physical construction	58%
Any finding involving administration	80%
Daily expenditure ledger not in accordance with procedures	50%
Procurement/tendering procedures not followed properly	38%
Insufficient documentation of receipt of materials	28%
Insufficient receipts for expenditures	17%
Receipts improperly archived	17%
Insufficient documentation of labor payments	4%

Note.—Tabulations from BPKP final report submitted to the Government of Indonesia's KDP management team and to the World Bank on December 22, 2004. This report included all findings from the 283 villages that were audited as part of phase II of the audits. The percentage reported is the percentage of the 283 audited village for which BPKP reported finding the listed problem.

(b) Table 7: audit findings

6.5 Table 8 and Table 9: nepotism and participation first stage

Read:

- Table 8 studies whether relatives of officials are more likely to work on audited projects.
- In audited villages, village-government family members are about 7.9 percentage points more likely to work on the project in column 1, and project-head family members are about 13.8 percentage points more likely in column 2.
- Table 9 confirms that the participation treatments increased grassroots participation.
- Invitations increased attendance by 14.8 people, and almost all of this increase came from nonelite villagers.
- Invitations also increased the number of people who spoke, including nonelite speakers.

Economic message: Audits may cause substitution toward less-monitored rents such as jobs for relatives. Meanwhile, the participation treatments succeeded at increasing meeting participation, so their limited corruption effects are not because the first stage failed.

TABLE 8
NEPOTISM

	(1)	(2)	(3)	(4)
Audit	−.011	.004	−.017	−.038
	(.023)	(.021)	(.032)	(.032)
Village government family member	−.020	.016	.016	−.014
	(.024)	(.017)	(.017)	(.023)
Project head family member	.051	−.015	.051	−.004
	(.032)	(.047)	(.032)	(.047)
Social activities	.017***	.017***	.013*	.014**
	(.006)	(.006)	(.006)	(.006)
Audit × village government family member	.079**			.064*
	(.034)			(.034)
Audit × project head family member		.138**		.115*
		(.060)		(.061)
Audit × social activities			.010	.008
			(.008)	(.008)
Stratum fixed effects	Yes	Yes	Yes	Yes
Observations	3,386	3,386	3,386	3,386
R^2	.26	.26	.26	.27
Mean dependent variable	.30	.30	.30	.30

NOTE.—Data are taken from the household survey. Each observation represents one household. Results come from estimating eq. (3), where the dependent variable is a dummy for whether a household member worked (for pay) on the road project. Estimation is by OLS with stratum fixed effects. Robust standard errors are in parentheses, adjusted for clustering at subdistrict level. Social activities refers to the number of social activities adult household members participated in during the last month. All specifications include controls for invitations and invitations plus comment

(a) Table 8: nepotism

TABLE 9
PARTICIPATION: FIRST STAGE

	Attendance (1)	Attendance of Nonelite (2)	Number Who Talk (3)	Number Nonelite Who Talk (4)
Invitations	14.83***	13.47***	.743***	.286***
	(1.35)	(1.25)	(.188)	(.079)
Invitations plus comments	11.48***	10.28***	.498***	.221***
	(1.35)	(1.27)	(.167)	(.069)
Meeting 2	−5.32***	−4.00***	.163	.024
	(1.11)	(1.06)	(.155)	(.084)
Meeting 3	−4.29***	−5.78***	.431**	−.158*
	(1.20)	(1.13)	(.172)	(.089)
Stratum fixed effects	Yes	Yes	Yes	Yes
Observations	1,775	1,775	1,775	1,775
R^2	.39	.38	.47	.28
Mean dependent variable	47.99	24.15	8.02	.94
p-value invitations = invitations + comment forms	.03	.03	.21	.43

NOTE.—Results come from estimating eq. (1), with the dependent variables the participation variables shown in the first row. Data are taken from the meeting survey. Each observation is a single village meeting. Stratum (subdistrict) fixed effects are included; since audit is constant within a subdistrict, the audit variable is automatically captured by the stratum fixed effect. Robust standard errors are in parentheses, adjusted for clustering at the village level.

(b) Table 9: participation first stage

6.6 Table 10 and Table 11A: meeting effects and participation effects on missing expenditures

Read:

- Table 10 shows that invitations and invitations plus comments increased the probability that a corruption-related problem was discussed by about 2.6 to 2.7 percentage points.
- Only invitations plus comments significantly increased the probability of a serious response being taken.
- Table 11A shows that the invitations treatment had little effect on total missing expenditures for road items or road plus ancillary projects.
- However, invitations reduced missing unskilled-labor expenditures by 18.7 percentage points without fixed effects and 21.5 percentage points with engineer fixed effects.

Economic message: Community participation can make meetings more informative and can protect worker wages, but it does not strongly reduce overall missing expenditures because most money is spent on materials.

TABLE 10
Participation: Impact on Meetings

	Number of Problems (1)	Any Corruption-Related Problem (2)	Serious Response Taken (3)
Invitations	.072	.027**	−.003
	(.063)	(.013)	(.008)
Invitations plus comments	.104	.026**	.015**
	(.064)	(.012)	(.008)
Meeting 2	−.187***	.002	−.020**
	(.066)	(.013)	(.009)
Meeting 3	−.428***	−.036***	−.029***
	(.074)	(.012)	(.009)
Stratum fixed effects	Yes	Yes	Yes
Observations	1,783	1,783	1,783
R^2	.50	.31	.22
Mean dependent variable	1.18	.07	.03
p-value invitations = invitations + comment forms	.60	.96	.02

Note.—Results come from estimating eq. (1), with the dependent variables the outcome of meetings shown in the first row. Data are taken from the meeting survey. Each observation represents one village. "Serious response" is defined as agreeing to replace a supplier or village office, agreeing that money should be returned, agreeing to an internal village investigation, asking for help from district project officials, or requesting an external audit. Estimation is by OLS.

(a) Table 10: participation and meeting content

TABLE 11
Participation: Main Theft Results

Percent Missing[a]	Control Mean (1)	Treatment Mean (2)	No Fixed Effects: Treatment Effect (3)	No Fixed Effects: p-Value (4)	Engineer Fixed Effects: Treatment Effect (5)	Engineer Fixed Effects: p-Value (6)	Stratum Fixed Effects: Treatment Effect (7)	Stratum Fixed Effects: p-Value (8)
	A. Invitations							
Major items in roads (N = 477)	.252	.230	−.021	.556	−.030	.385	−.026	.448
	(.033)	(.033)	(.035)		(.034)		(.034)	
Major items in roads and ancillary projects (N = 538)	.268	.236	−.030	.360	−.032	.319	−.029	.356
	(.031)	(.031)	(.032)		(.032)		(.032)	
Breakdown of roads:								
Materials (N = 477)	.209	.221	.014	.725	.008	.839	.005	.882
	(.041)	(.041)	(.038)		(.037)		(.037)	
Unskilled labor (N = 426)	.369	.180	−.187*	.058	−.215**	.024	−.143*	.098
	(.077)	(.077)	(.098)		(.094)		(.086)	

(b) Table 11A: invitations and missing expenditures

6.7 Table 11B and Table 12A: comment forms and distribution mechanisms

Read:

- Table 11B shows that invitations plus comment forms have small and statistically insignificant average effects on overall missing expenditures.
- The labor effect is negative but smaller and less precisely estimated than the invitations-only labor effect.
- Table 12A studies whether the way invitations were distributed matters.
- Invitations distributed via neighborhood heads or via schools do not produce large statistically significant reductions in total missing expenditures.

Economic message: Anonymous communication alone is not enough. The design of who controls information distribution matters because local elites can shape who receives monitoring tools.

Percent Missing[a]	Control Mean (1)	Treatment Mean (2)	No Fixed Effects: Treatment Effect (3)	No Fixed Effects: p-Value (4)	Engineer Fixed Effects: Treatment Effect (5)	Engineer Fixed Effects: p-Value (6)	Stratum Fixed Effects: Treatment Effect (7)	Stratum Fixed Effects: p-Value (8)
	B. Invitations Plus Comments							
Major items in roads (N = 477)	.252	.228	−.022	.455	−.024	.411	−.015	.601
	(.033)	(.026)	(.030)		(.029)		(.030)	
Major items in roads and ancillary projects (N = 538)	.268	.238	−.026	.409	−.025	.406	−.027	.385
	(.031)	(.026)	(.032)		(.030)		(.031)	
Breakdown of roads:								
Materials (N = 477)	.209	.180	−.028	.414	−.022	.496	−.010	.754
	(.041)	(.032)	(.034)		(.032)		(.033)	
Unskilled labor (N = 426)	.369	.267	−.099	.255	−.132	.131	−.090	.323
	(.077)	(.073)	(.087)		(.087)		(.091)	

Note.—See the note to table 4. Results come from estimating eq. (1), a regression of the dependent variable on a dummy for audit treatment, invitations treatment, and invitations plus comment forms treatments. Each invitations effect and invitations plus comments effect comes from a separate regression, with the dependent variable listed in the row and the fixed effects specification listed in the column heading. Robust standard errors are in parentheses. Regressions without stratum (i.e., subdistrict) fixed effects include a variable for audits and allow for clustering of standard errors by subdistrict.

[a] Percent missing equals log reported value − log actual value.

* Significant at 10 percent.

** Significant at 5 percent.

*** Significant at 1 percent.

(a) Table 11B: invitations plus comment forms

TABLE 12
Interactions of Participation Experiments with How Invitations Were Distributed

Percent Missing[a]	Control Mean (1)	Treatment Mean (2)	No Fixed Effects: Treatment Effect (3)	No Fixed Effects: p-Value (4)	Engineer Fixed Effects: Treatment Effect (5)	Engineer Fixed Effects: p-Value (6)	Stratum Fixed Effects: Treatment Effect (7)	Stratum Fixed Effects: p-Value (8)
	A. Invitations							
	Invitations Distributed via Neighborhood Heads							
Major items in roads (N = 246)	.252	.222	−.030	.469	−.043	.274	−.042	.324
	(.033)	(.044)	(.042)		(.039)		(.043)	
Major items in roads and ancillary projects (N = 271)	.268	.255	−.013	.761	−.015	.712	−.004	.924
	(.031)	(.045)	(.043)		(.041)		(.043)	
	Invitations Distributed via Schools							
Major items in roads (N = 233)	.252	.239	−.009	.854	−.014	.774	−.003	.950
	(.033)	(.046)	(.050)		(.048)		(.045)	
Major items in roads and ancillary projects (N = 263)	.268	.216	−.048	.282	−.051	.245	−.056	.155
	(.031)	(.040)	(.044)		(.043)		(.039)	

(b) Table 12A: invitations by distribution method

6.8 Table 12B and Table 13: elite capture and cost-benefit analysis

Read:

- Table 12B shows the important result for comment forms: when invitations plus comment forms were distributed through schools, missing expenditures fell substantially.
- When comment forms were distributed by neighborhood heads, the treatment effect is not negative and is sometimes positive.

- Table 13 performs a cost-benefit analysis of the audit treatment.
- Equal-weighted net benefits are estimated at US$245 per village, and distribution-weighted net benefits are US$508 per village.

Economic message: Grassroots monitoring is vulnerable to elite capture, but it can work when information channels bypass local officials. Audits appear cost-effective even after accounting for administrative and time costs.

	B. Invitations Plus Comments Invitations Plus Comment Forms Distributed via Neighborhood Heads							
Major items in roads (N = 242)	.252	.278	.025	.483	.038	.294	.022	.602
	(.033)	(.036)	(.036)		(.036)		(.041)	
Major items in roads and ancillary projects (N = 271)	.268	.277	.010	.792	.024	.535	.023	.569
	(.031)	(.039)	(.039)		(.038)		(.040)	
	Invitations Plus Comment Forms Distributed via Schools							
Major items in roads (N = 242)	.252	.179	−.070*	.093	−.086**	.023	−.052	.150
	(.033)	(.036)	(.041)		(.038)		(.036)	
Major items in roads and ancillary projects (N = 267)	.268	.198	−.064	.127	−.077*	.052	−.078*	.056
	(.031)	(.034)	(.042)		(.039)		(.041)	

NOTE.—See the note to table 11. N refers to the number of observations with nonmissing dependent variable for control observations (i.e., where neither invitations nor invitations and comment forms were distributed) plus the number of treatment observations for the listed treatment. Treatment effects and p-values are computed from a regression of the dependent variable on a dummy for audit treatment and four dummies for the participation treatments interacted with distribution mechanism (i.e., invitations distributed via neighborhood heads, invitations distributed via schools, invitations plus comment forms distributed via neighborhood heads, and invitations plus comment forms distributed via schools).
[a] Percent missing equals log reported value − log actual value.
* Significant at 10 percent.
** Significant at 5 percent.
*** Significant at 1 percent.

(a) Table 12B: comment forms by distribution method

TABLE 13
COST-BENEFIT ANALYSIS OF AUDIT TREATMENT

	Equal-Weighted Net Benefits (1)	Distribution-Weighted Net Benefits (2)
Cost of treatment:		
Monetary cost	−335	−278
Associated deadweight loss	−134	−111
Time cost	−31	−31
Subtotal	−500	−419
Change in rents received by corrupt officials:		
From theft of materials	−367	−224
From theft of wages	−102	−62
Subtotal	−468	−286
Change in benefits from project:		
Net present value of road expenditures	1,165	1,165
Wages received by workers	86	86
Other expenditures	−37	−37
Subtotal	1,213	1,238
Total net benefits	245	508

NOTE.—All figures are given in U.S. dollars. Costs are listed as negative numbers. Distributional weights were calculated

(b) Table 13: cost-benefit analysis of audits

6.9 Table B1: calibration assumptions

Read:

- Table B1 reports loss ratios and worker-capacity assumptions used to translate engineering measurements into estimated actual costs.
- The loss ratios are 1.00 for sand, 1.20 for rocks, and 1.75 for gravel.
- Worker-capacity assumptions include spreading sand, splitting rocks, installing rock layers, spreading gravel, digging side channels, and building retaining walls.
- These assumptions come from calibration exercises and KDP engineering assumptions.

Economic message: The calibration table is crucial because the corruption measure depends on translating physical road measurements into dollar costs. The paper explicitly addresses the engineering assumptions behind the main outcome variable.

TABLE B1
Assumptions for Loss Ratios and Worker Capacity

	Results from Calibration
Loss ratios:	
Sand	1.00
Rocks	1.20
Gravel	1.75
Worker capacity:	
Clearing brush and cleaning road surface (m^2)	20
Spreading sand (m^3)	4.5
Splitting rocks (m^3)	3.0
Installing rock layer (m^2)	6.5
Spreading gravel (m^3)	2.25
Digging side channels and creating shoulder (m^3)	1.0
Building retaining wall (m^3) (unskilled labor portion)	.33*
Standard cut and fill (m^3)	2*

Note.—All loss ratios are expressed as the ratio of the original amount of material to the amount measured and pertain to measurements loose, not compacted. Worker capacity is the quantity of the given activity that can be done by one person per six hours of work.
* Original KDP assumption that was not able to be reconfirmed from calibration exercises.

Table B1: assumptions for loss ratios and worker capacity

7 Short summary

- Olken studies corruption in 608 Indonesian village road projects using a randomized field experiment.
- The paper compares audits, meeting invitations, and anonymous comment forms.
- Corruption is measured as missing expenditures: log reported cost minus log independently estimated actual cost.
- Missing expenditures average roughly one-quarter of project spending.
- Increasing the probability of government audits from about 4% to 100% reduces missing expenditures by about eight percentage points.
- The audit effect comes mainly through quantities rather than prices.
- Audits do not eliminate missing expenditures because audit findings are often procedural and sanctions are imperfect.
- Audits may increase nepotistic allocation of paid project jobs to relatives of officials.
- Invitations increase attendance and nonelite participation at accountability meetings.
- Grassroots participation has little average effect on overall missing expenditures but reduces missing labor expenditures.
- Comment forms reduce missing expenditures only when distributed through schools, bypassing village officials.
- The audit intervention appears cost-effective in the paper's cost-benefit analysis.

8 Conclusion

8.1 Core conclusion

Olken shows that corruption in local public works projects can be reduced by external audits. Increasing the audit probability from a low baseline to near certainty produced a sizable fall in missing expenditures. By contrast, increasing grassroots participation had weaker average effects and worked mainly in specific situations where private incentives to monitor were stronger or elite capture was limited.

8.2 Economic intuition

The economic intuition is a monitoring-incentives argument. Local officials and project teams can extract rents by overstating materials or labor. Audits raise the expected cost of doing so because outside monitors may identify irregularities and publicly reveal them. Grassroots monitors have better local information but weaker incentives when the benefit of monitoring is a public good. Workers care directly about missing wages, so labor theft is more responsive to participation than materials theft.

8.3 Incremental contribution revisited

The paper advances the empirical corruption literature in three main ways. First, it measures corruption directly using engineering estimates rather than perceptions. Second, it uses randomized treatments to identify causal effects of anticorruption policies. Third, it reveals why different monitoring systems work differently: audits change expected punishment, while grassroots monitoring depends on incentives, information, and control over the monitoring process.

8.4 Implications for policy

The findings support the use of professional audits even in environments where corruption is widespread. They also caution against treating community monitoring as a universal substitute for formal enforcement. Community monitoring is more likely to work when beneficiaries have a direct private stake in monitoring and when information channels cannot be captured by local elites.

8.5 Limitations

The experiment measures short-run effects in a specific Indonesian road-building program. The missing-expenditure measure is carefully constructed but still depends on engineering assumptions. The audit treatment changes the probability of being audited but not necessarily the probability of formal punishment. The paper also cannot fully assess long-run adaptation, such as repeated relationships with auditors, changes in who becomes a village official, or future shifts to harder-to-detect forms of corruption.

8.6 Takeaway

The enduring takeaway is that anticorruption policy depends on both information and incentives. Top-down audits can meaningfully reduce theft, while bottom-up monitoring works only when citizens have the incentive and independence to act on information. In this setting, professional external monitoring was the more reliable tool for reducing corruption in public infrastructure.